

SmartHR AI - Employee Attrition Prediction System

A full-stack web application that predicts employee attrition using machine learning concepts, built with Spring Boot, MySQL, HTML/CSS/JS, and Bootstrap.

Features

- **Employee Attrition Prediction** - Rule-based scoring algorithm evaluating 10+ employee attributes
- **Interactive Dashboard** - Real-time HR statistics with charts (Chart.js)
- **Prediction History** - Track all predictions with timestamps and confidence scores
- **Reports & Analytics** - Attrition distribution, department analysis, income charts
- **User Authentication** - Register/Login with session management
- **Dark Mode** - Full dark theme across all pages with settings persistence
- **Responsive Design** - Works on desktop and tablet screens
- **Configurable Settings** - General, notifications, and appearance preferences

Tech Stack

Layer	Technology
Backend	Spring Boot 3.5, Java 17, Spring Data JPA
Database	MySQL 8.0 (via XAMPP)
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5.3.3
Charts	Chart.js
Build	Maven
Dataset Import	Python 3, mysql-connector

Project Structure

```
SmartHR-AI/
├── backend/                                # Spring Boot application
│   ├── src/main/java/com/smarthr/smarthr_ai/
│   │   ├── SmarthrAiApplication.java
│   │   ├── config/CorsConfig.java
│   │   ├── controller/                  # REST API controllers
│   │   ├── service/                    # Business logic
│   │   ├── entity/                     # JPA entities
│   │   ├── repository/                 # Data access
│   │   ├── dto/                        # Request/Response DTOs
│   │   └── exception/                  # Error handling
│   └── pom.xml
├── frontend/                              # Static web files
│   ├── *.html                          # 7 HTML pages
│   ├── css/                            # Stylesheets
│   └── js/                             # JavaScript modules
├── dataset/                              # IBM HR Analytics CSV
├── ml/                                   # ML training scripts (Python)
├── models/                              # Saved ML models
├── import_dataset.py                     # Dataset import script
└── README.md
```

Prerequisites

- Java JDK 17+
- Apache Maven 3.8+
- MySQL 8.0 (via XAMPP)
- Python 3.8+ (for dataset import)
- Modern web browser

Setup

1. Start MySQL

Start Apache and MySQL modules from XAMPP Control Panel.

2. Create Database

```
CREATE DATABASE smarthr_ai;
```

3. Import Dataset

```
pip install mysql-connector-python
python import_dataset.py
```

4. Build Backend

```
cd backend
mvn clean install
```

5. Run Backend

```
mvn spring-boot:run
```

Backend runs on `http://localhost:8080`.

6. Open Frontend

Open `frontend/login.html` in your browser, or serve with Live Server.

API Endpoints

Method	Endpoint	Description
POST	<code>/api/auth/register</code>	Register new user
POST	<code>/api/auth/login</code>	User login
GET	<code>/api/dashboard/stats</code>	Dashboard statistics
POST	<code>/api/predictions/predict?userId=</code>	Make a prediction
GET	<code>/api/predictions/history/{userId}</code>	Get prediction history

Usage

1. **Register** a new account on the registration page
2. **Login** with your credentials
3. **Dashboard** shows employee stats, charts, and insights
4. **Prediction** - Fill the form with employee details to predict attrition risk
5. **History** - View all past predictions
6. **Reports** - Explore interactive analytics charts
7. **Settings** - Customize theme, notifications, and appearance

Dataset

Uses the [IBM HR Analytics Employee Attrition Dataset](#) with:

- 1,470 employee records
- 31 features (age, department, income, satisfaction, etc.)
- 16.1% attrition rate (237 left, 1,233 stayed)

License

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